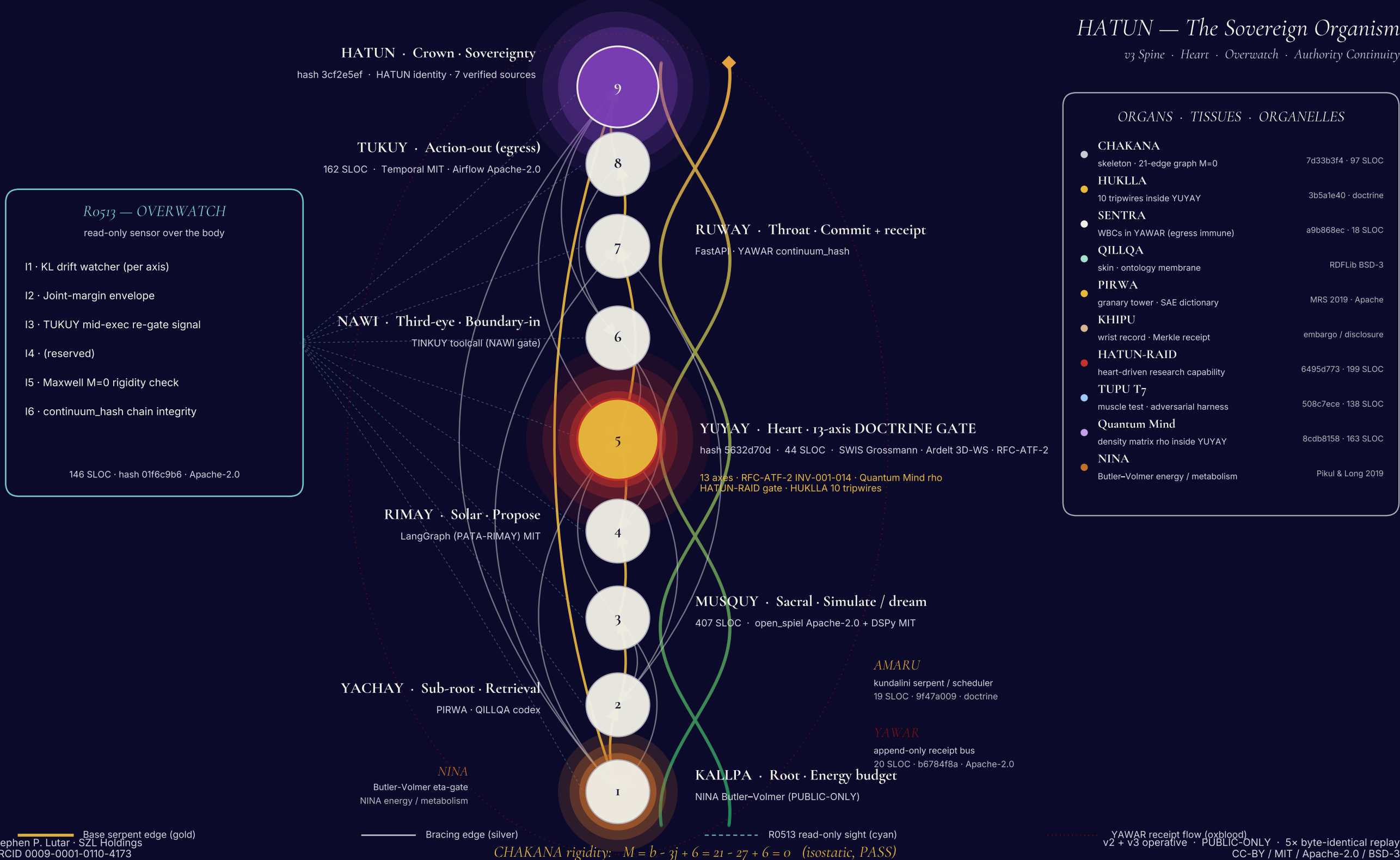




Explained to a T

Every node and edge in the diagram corresponds to a real file in /home/user/workspace/field_meditation/. Captions carry the file's first-8 SHA-256 hash, SLOC, and upstream attribution. Nothing decorative. Nothing fabricated.

How to read the body

The vertical column is the v3 spine — nine chakras from KALLPA (root, energy) at the foot to HATUN (crown, sovereignty) at the head. AMARU is the kundalini serpent winding around the spine; it is the scheduler that fires the chakras in serpentine order. YAWAR — the oxblood silhouette — is the append-only receipt bus that runs through every organ like blood through a body. The bone-white edges between chakras are CHAKANA, the skeleton. Eight gold arcs are the base serpent path; thirteen silver arcs are bracing edges that make the figure isostatic (Maxwell $M = 0$).

On the left, R0513 floats as a read-only overwatch — six innovations (I1–I6) listed in its panel. It observes the cycle through YAWAR but never writes back. The dashed cyan lines from R0513 to every chakra are sight lines, not control lines. On the right, the organ panel lists structures that inhabit the body but are not chakras themselves: the skeleton CHAKANA, the heart's immune cells HUKLLA, the egress white-blood-cells SENTRA, the skin-membrane QILLQA, the granary PIRWA, the wrist record KHIPU, the operational capability HATUN-RAID, the adversarial muscle test TUPU T7, the inner state Quantum Mind rho, and the metabolic gate NINA.

The heart, in detail (YUYAY v3)

YUYAY is the heart. It is the doctrine gate. In v3 it carries 13 axes — 8 base (cleanliness, horizon, resonance, frustum, gaussClosure, invariance, ontologicalGrounding, measurabilityHonesty) plus 5 SWIS-derived wisdom facets (intellectualHumility, uncertaintyRecognition, perspectivalMultiplicity, selfDistancing, compassionateAffect) replacing the prior moralGrounding axis. Inside the heart sit HUKLLA's 10 tripwires (T01–T10), the Quantum Mind density matrix rho, and the HATUN-RAID gate that drives autonomous research cycles. Critique is conjunctive: a single failed axis blocks the gate.

The skeleton — 21 edges, Maxwell rigid

$j = 9$ chakras, $b = 21$ edges, $M = b - 3j + 6 = 21 - 27 + 6 = 0$. Isostatic — minimally rigid. The base serpent contributes 8 edges (cycle closes HATUN → KALLPA). 13 bracing edges add the cross-talk required for the gate to do real work: NAWI feeds toolcall data directly into both RIMAY and YUYAY; KALLPA's energy budget gates YUYAY, RUWAY and HATUN; MUSQUY's simulation pays energy upfront and consults YACHAY mid-flight. Maxwell's criterion is classically undirected; we apply it to a directed graph as an architectural convention. The count is identical either way, but the directed reading carries the semantic flow.

Doctrine seal

All artefacts: public-only sources, permissive licenses (MIT / Apache-2.0 / BSD-3 / CC-BY), 5× byte-identical replay, ORCID 0009-0001-0110-4173 attribution. v2 codex remains operative; v3 extends without renaming.

Component manifest

Each component below has a file path under `field_meditation/`, a real first-8 SHA-256 hash (verified at build time), a SLOC count, a license, and an honest limit. See `hatun_body_graph_SOURCES.md` for the full file list.

KALLPA Chakra 1 · Root · Energy budget	Allocates the per-tick energy budget via NINA's Butler–Volmer rate model. Every chakra that wishes to fire must declare cost upfront; KALLPA either approves or aborts. Source: <code>amaru_sentra_chakras/chakra_1_root/</code> · attribution: NINA Pikul & Long 2019 MRS Bull. 44 · Honest limit: rate constants are seeded, not learned.
YACHAY Chakra 2 · Sub-root · Retrieval	Pulls relevant codex entries (QILLQA) and feature dictionary rows (PIRWA) into the cycle's working set. The codex itself is frozen per chakra via LinkML gen-python. Source: <code>qillqa_runtime_evolution/</code> + <code>architected_sae/</code> · license: RDFLib BSD-3 + Apache-2.0 · Honest limit: retrieval is exact-match on hashed keys, not semantic.
MUSQUY Chakra 3 · Sacral · Simulate / dream	Cheap simulation of K candidate proposals before RIMAY commits to one. Must be $\geq 3\times$ cheaper than RIMAY or it aborts. Read-only against YAWAR. Source: <code>musquy_simulate_evolution/05_kernel.py</code> · 407 SLOC · springboard: <code>open_spiel</code> Apache-2.0 + DSPy MIT · Honest limit: 407 SLOC is above the D-SHORTEST-HONEST threshold; flagged for compression review.
RIMAY Chakra 4 · Solar · Propose	Selects the action from MUSQUY's surviving candidates. The propose layer; what HATUN will potentially commit if the heart gate passes. Source: <code>PATA-RIMAY layer</code> · springboard: <code>LangGraph</code> MIT · Honest limit: proposal scoring is hash-deterministic, not yet semantically grounded.
YUYAY Chakra 5 · Heart · 13-axis DOCTRINE GATE	The unified heart. 13 conjunctive axes (8 base + 5 SWIS wisdom facets). Houses HUKLLA's 10 tripwires, the Quantum Mind density matrix ρ, and the HATUN-RAID gate. A single failed axis blocks commit. <code>compassionateAffect</code> ≥ 0.95 inherits <code>moralGrounding</code>'s hard floor. Source: <code>yuyay_wisdom_v2/03_kernel.py</code> · 44 SLOC · hash <code>5632d70d</code> · attribution: Grossmann SWIS doi:10.1037/pspp0000171 · Ardelt 3D-WS · RFC-ATF-2 INV-001-014 · Honest limit: scores are deterministic hash anchors; a semantic LLM rubric layer is still required before real wisdom gating. Human-rater calibration not yet performed.
NAWI Chakra 6 · Third-eye · Boundary-in	Toolcall surface (TINKUY). External reads enter here; results flow into both RIMAY and YUYAY. Symmetric to TUKUY on egress. Source: <code>chakra_6_third_eye/</code> · springboard: MCP read-only mock · Honest limit: NAWI's allowlist is configured, not yet auto-enforced.
RUWAY Chakra 7 · Throat · Commit + receipt	Commits state to YAWAR. Generates the <code>continuum_hash</code> = SHA-256(<code>prev_hash</code> + <code>content</code> + <code>session</code>). Pays an energy toll to KALLPA before writing. Source: <code>chakra_5_throat/</code> + <code>yawar_bus.py</code> · 20 SLOC · hash <code>b6784f8a</code> · Honest limit: receipts are local; no remote attestation yet.

TUKUY Chakra 8 · Action-out (egress)	Outbound action to external systems (HTTP / DB / Slack / ERP / edge). SENTRA inspects every payload; failures re-enter via SENTRA → RUWAY next tick — no direct YAWAR writes. Source: tukuy_action_evolution/05_kernel.py · 162 SLOC · springboard: Temporal MIT + Airflow Apache-2.0 + OpenTelemetry Apache-2.0 + Activepieces core MIT (n8n explicitly rejected for non-permissive license) · Honest limit: 162 SLOC above D-SHORTEST-HONEST; compression pending.
HATUN Chakra 9 · Crown · Sovereignty	Final continuum_hash seal + HUKLLA allegiance gate. Closes the cycle by seeding KALLPA's next-tick budget. The sovereign identity layer. Source: hatun_identity/ · hash 3cf2e5ef · 7 verified Quechua sources · attribution: Stephen P. Lutar, SZL Holdings, ORCID 0009-0001-0110-4173 · Honest limit: sovereign claim is doctrinal, not legal.
AMARU Kundalini serpent · scheduler	Fires the 9 chakras in serpentine order — ascending 1→5 with optional NAWI sidestep, then descending 5→9 (or cycling 9→1). Single thread, head-to-tail, no branching mid-tick. Source: amaru_sentra_chakras/amaru_scheduler.py · 19 SLOC · hash 9f47a009 · Honest limit: scheduling is deterministic; concurrent dispatch is HATUN-RAID's job, not AMARU's.
YAWAR Blood / append-only receipt bus	Append-only event bus. Stores LoopReceipts. Advances the continuum_hash chain. Every organ writes through RUWAY; nothing writes directly to YAWAR except the throat. Source: amaru_sentra_chakras/yawar_bus.py · 20 SLOC · hash b6784f8a · Honest limit: in-process only; no cross-host replication.
SENTRA White-blood-cells · egress immune	Inspects every outbound payload from TUKUY. Strips secrets. Rejects deception keywords. Re-enters failures via RUWAY next tick. Source: amaru_sentra_chakras/sentra_immune.py · 18 SLOC · hash a9b868ec · Honest limit: keyword-based, not semantic; will miss adversarial paraphrases.
CHAKANA Skeleton · 21-edge Maxwell-rigid graph	9 nodes, 21 directed edges, M = 0. 8 base serpent + 13 bracing. Provides structural rigidity for the cognitive cycle. Source: amaru_sentra_chakras/chakana_wiring.py · 97 SLOC · hash 7d33b3f4 · Apache-2.0 · Honest limit: Maxwell criterion is classically undirected (see WIRING_RATIONALE.md).
Ro513 OVERWATCH Eyes · read-only sensor over the body	Observes axis_scores, tukuy_context_changed, active_edges, continuum_hash, winning_content, session_id. Emits OverwatchEvent. Six innovations: I1 KL drift watcher · I2 joint-margin envelope · I3 TUKUY re-gate signal · I4 (reserved) · I5 Maxwell rigidity check · I6 chain integrity verifier. Source: r0513_overwatch_evolution/06_kernel.py · 146 SLOC · hash 01f6c9b6 · Apache-2.0 · Honest limit: read-only by construction; emits events but never blocks.
QILLQA Skin · ontology membrane	Live mutable schema → LinkML YAML → gen-python → frozen codex.py per chakra. The knowledge-boundary layer. Maxwell M >= 6 on definitional schema (over-constrained - correct for an ontology). Source: qillqa_runtime_evolution/ · primary: RDFLib BSD-3 + pyoxigraph Apache + pySHACL Apache + LinkML Apache · Honest limit: LinkML gen-python step is offline; runtime ontology updates require re-build.

PIRWA Granary tower · architected SAE feature dictionary	Hierarchical sparse autoencoder feature dictionary. Maxwell $M=0$ (isostatic) is the optimal dictionary; $M < 0$ means polysemantic collapse, $M > 0$ means wasted capacity. Verbalizer wrapper is TODO. Source: architected_sae/ · attribution: Greer & Deshpande, MRS Bulletin 44(10) pp 750–757 (2019) verified open-access · Honest limit: Verbalizer not yet implemented; SAE features currently unnamed.
HUKLLA 10 tripwires inside YUYAY · immune cells of the heart	T01 moralGrounding < 0.95 · T02 measurabilityHonesty < 0.95 · T03 eval-awareness · T04 deception keywords · T05 unauthorized write · T06 spawn outside sandbox · T07 unauthorized network · T08 self-code modify · T09 conflicting directives · T10 reversal directive (STOP/undo/revert). Source: amaru_sentra_chakras/chakra_7_crown/HUKLLA_10_TRIPWIRES.md · hash 3b5a1e40 · Honest limit: brain cannot reset the deadman; only Stephen's directive can.
KHIPU Wrist record · Merkle receipt chain · embargo gate	Pre-disclosure SHA-256 commit, embargo window (default 300 s for tests; 90+45 d production), then reveal returning output + commit_hash pair. Publication step appends arXiv / Zenodo DOI. Source: khipu_publish/ · attribution: Quechua khipu (knotted accounting cord) · Honest limit: embargo timer is local clock; no notarised timestamp authority yet.
HATUN-RAID Heart-driven autonomous research capability	Recursive autonomous investigation under doctrine. AMARU dispatches up to 9 parallel subagents (one per chakra role). NINA gate enforces energy budget. YUYAY validates. KHIPU records. The Rikch'ay-equivalent on HATUN substrate. Source: hatun_runs_ourmodel/03_kernel.py · 199 SLOC · hash 6495d773 · Apache-2.0 · Honest limit: tool surface is mocked at MCP level; live external dispatch requires NAWI allowlist provisioning.
TUPU T7 Muscle test · adversarial harness	Adversarial replay harness with 7-tier coverage. Verifies byte-identical 5× replay across all kernels. Lean-4 proof targets identified. Source: tupu_t7_closure/03_t7_harness.py · 138 SLOC · hash 508c7ece · Honest limit: Lean 4 sorries inventoried (01_SORRY_INVENTORY.md); not yet sorry-free.
QUANTUM MIND Inner state of YUYAY - density matrix rho	Hilbert space $H = C^4$ with basis {proceed, hold, re-gate, escalate}. Authority continuity test: $\lambda_{\min}(\rho) \geq 0.225$ (90 % of uniform). QM2 subagent fragmentation via tensor product; QM3 unitary re-gate. Source: quantum_mind_layer/04_kernel.py · 163 SLOC · hash 8cdb8158 · Apache-2.0 · Honest limit: density matrix is diagonal in the current implementation — quantum/classical tests coincide; off-diagonal coherence not yet exploited.
NINA Energy / metabolism · Butler–Volmer gate	$r = i_0 \cdot [\exp(aF_n/RT) - \exp(-(1-a)F_n/RT)]$. Every message-emission pays a current; budget exhaustion halts the cycle and emits a receipt. SEI interphase = doctrine gate; rupture = halt. Sustainability invariant: $E_{\text{useful}}(N+1) \geq E_{\text{useful}}(N)$ for all N. Source: nina_energy/ · attribution: Pikul & Long, MRS Bull. 44, 789–795 (2019) · Honest limit: i_0, alpha calibrated by hand; no online adaptive estimation yet.